

Football Copilot - Development Roadmap (v1)

Final Tech Stack

Frontend: React 19, Vite, JavaScript, Tailwind CSS v4, shadcn/ui, React Router v7, Axios, TanStack Query, React Hook Form, Zod, Framer Motion, Lucide React, TanStack Table, Recharts, Sonner, dayjs, React Dropzone, React Context + useState + useReducer.

Backend: Node.js, Express, MongoDB, Redis, BullMQ, Socket.IO.

AI: OpenAI/Gemini, LangChain, LangGraph, Embeddings, Qdrant/pgvector.

Deployment: Vercel, Railway, Cloudinary.

Removed: Zustand, Redux, TypeScript.

Phase 1 - Football Platform

Sprint 1: Authentication (Login, Register, Refresh Token, Forgot Password, Profile, Avatar, optional Google Login).

Sprint 2: Dashboard (Today, Live, Upcoming, Yesterday).

Sprint 3: Competitions (EPL, La Liga, Serie A, Bundesliga, Ligue 1, UCL).

Sprint 4: Match Details (Timeline, Stats, Lineups, Events, xG, Cards, Goals).

Sprint 5: Team Pages (Fixtures, Form, Standings, Coach, Injuries).

Sprint 6: Player Pages (Stats, Career, Transfers).

Sprint 7: News aggregation into MongoDB.

Sprint 8: User features (Favorites, Bookmarks, Notifications, Dark Theme).

Sprint 9: Scheduler (Cron jobs for matches, standings, news, stats).

Sprint 10: Deploy Football Platform MVP.

Phase 2 - AI Features

Sprint 11: LLM Chat.

Sprint 12: AI Match Summary.

Sprint 13: AI News Summary.

Sprint 14: Explain Football Concepts & Matches.

Sprint 15: Tool Calling (matches, standings, players, transfers, stats).

Sprint 16: AI Recommendations.

Sprint 17: Intelligent Notifications.

Deploy AI MVP.

Phase 3 - AI Engineering

Sprint 18: RAG.

Sprint 19: Embeddings.

Sprint 20: Vector Database (Qdrant/pgvector).

Sprint 21: Memory.

Sprint 22: Planner Agent.

Sprint 23: Multi-Agent System.

Sprint 24: AI Timeline.

Sprint 25: AI Scout.

Sprint 26: AI Debate.

Sprint 27: AI Daily Briefing.

Sprint 28: Voice Assistant.

Sprint 29: Streaming Responses.

Sprint 30: Final Deployment.

Architecture Principles

Every phase is deployable. Backend written from scratch. UI generated/refined with AI tools but architecture and business logic remain handcrafted. Think of the product as an AI Operating System for Football, not just a football website with chat.